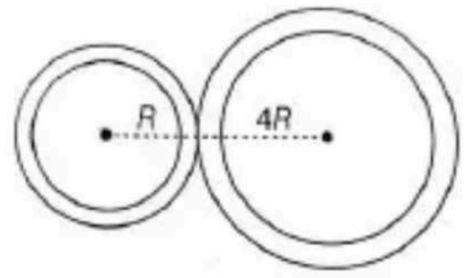
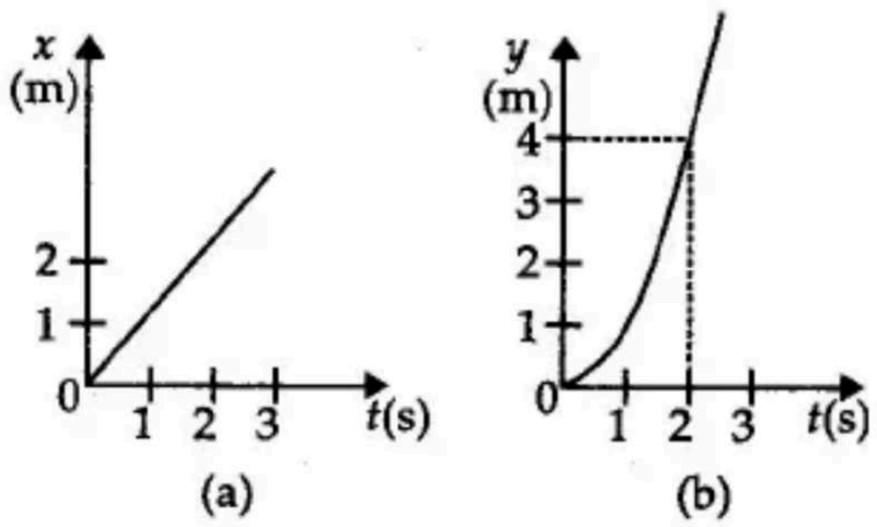
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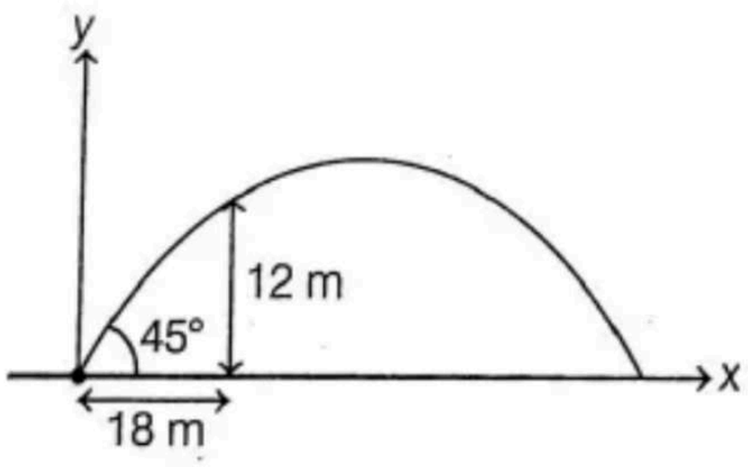

R
4R

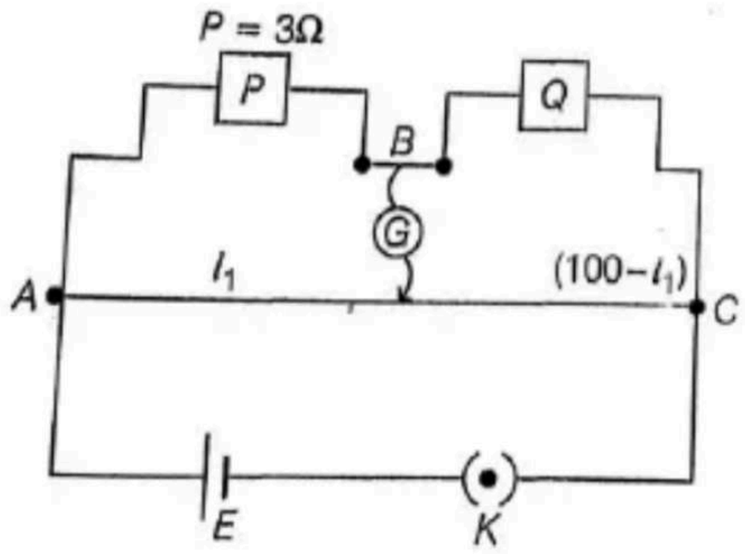
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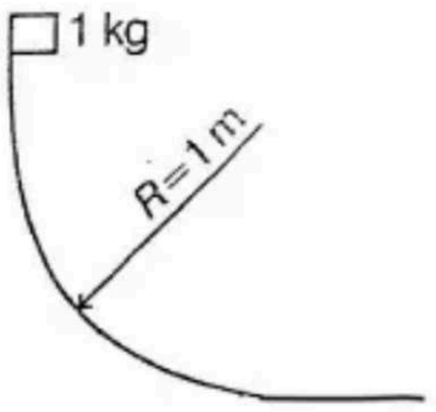
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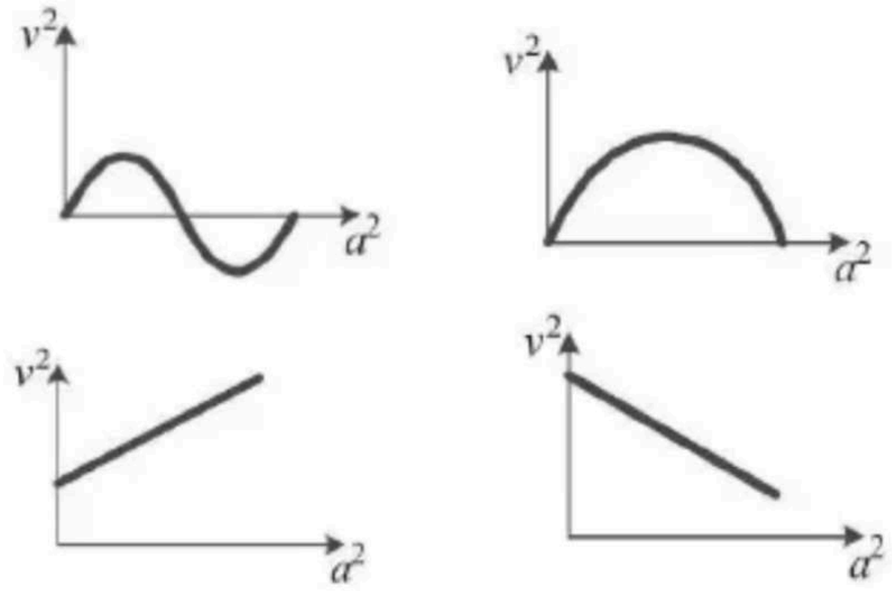
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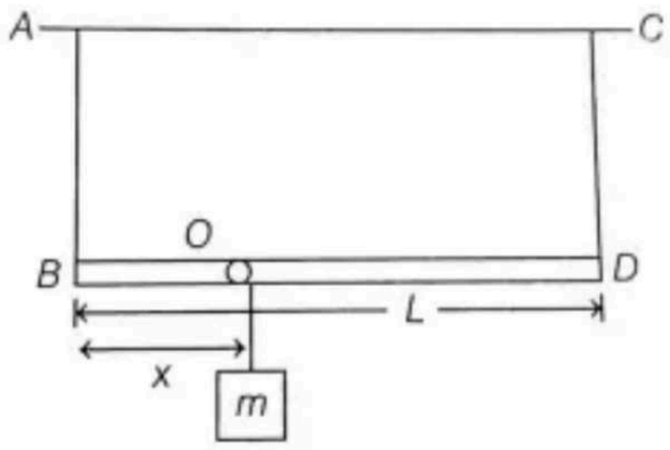

x
(m)
2
1
0
1 2 3
t(s)
(a)
y
(m)
4
3
2
1
0
1 2 3
t(s)
(b)

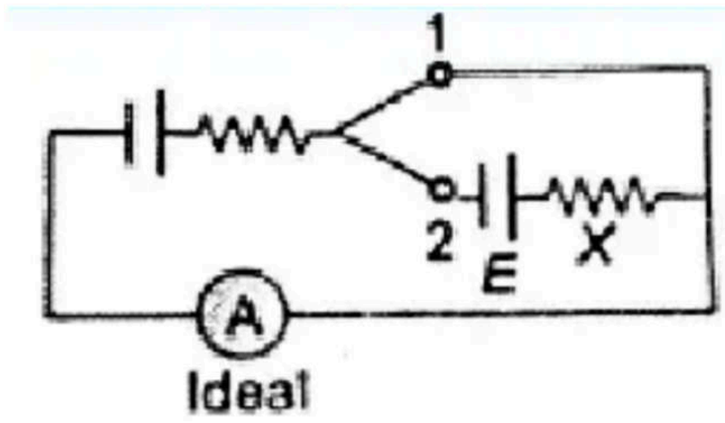

y
12 m
45°
x
18 m

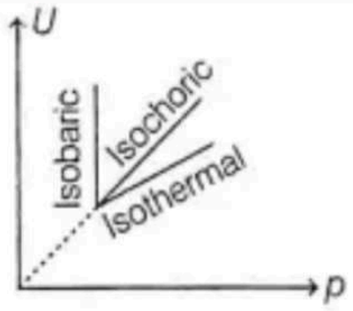

P = 3Ω
P
Q
B
G
A
l_1
$(100-l_1)$
C
E
K

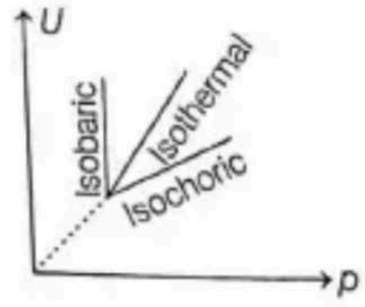

1 kg
R=1 m

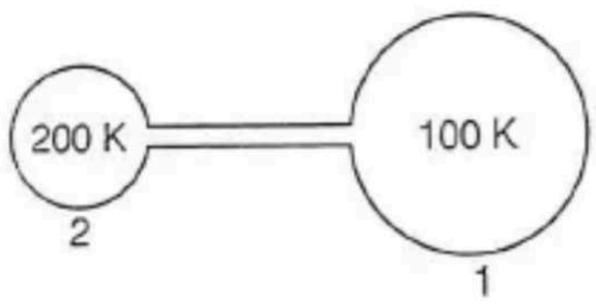

v^2
a^2
v^2
a^2
v^2
a^2
v^2
a^2

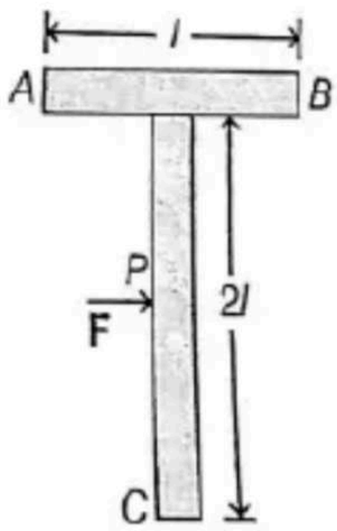

A
C
O
B
D
L
x
m

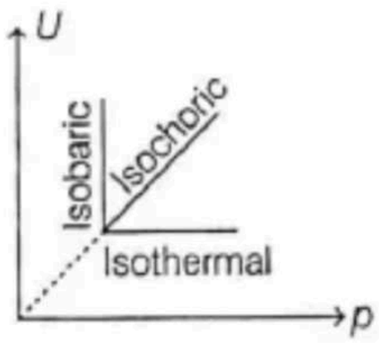

1
2
E
X
A
Ideal

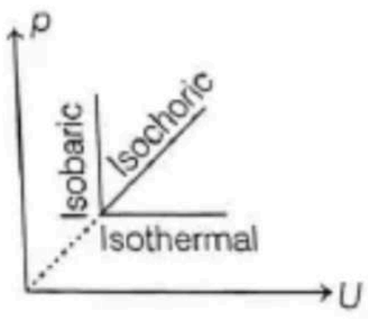

U
Isobaric
Isochoric
Isothermal
p


U
Isobaric
Isothermal
Isochoric
p


200 K
100 K
2
1


l
A
B
P
F
2l
C


U
Isobaric
Isochoric
Isothermal
p


p
Isobaric
Isochoric
Isothermal
U

$\frac{\lambda \times \lambda}{\lambda - \lambda}$

$\frac{\lambda \times \lambda}{\lambda + \lambda}$

$\frac{\lambda \times \lambda}{\lambda - \lambda}$

$\frac{h}{h}$

$\frac{h}{h}$

$\frac{\pi}{\pi}$

$\frac{\pi}{\pi}$

A	B	X
0	0	0
0	1	1
1	1	0

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

Wb/m

Wb/m

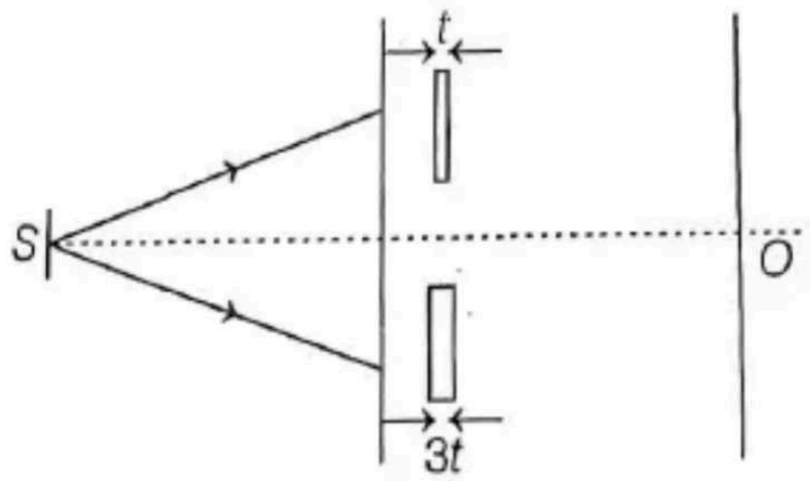
Wb/m

Wb/m

Wb/m

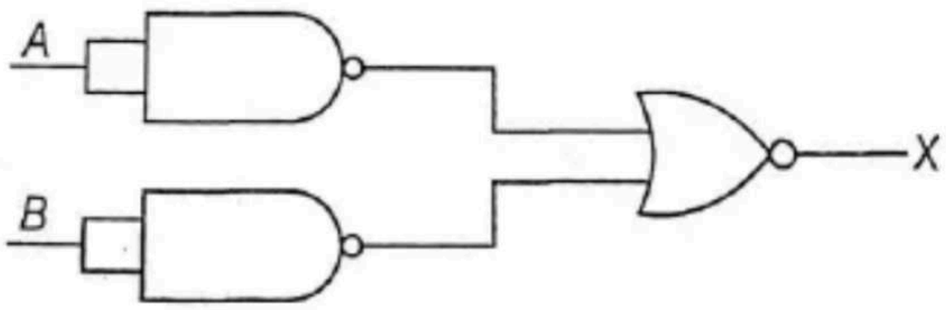
l

λ



μt μ t

μ t μt



49. Given below are two statements:

Statement-A:For halogens ionisation enthalphies, electronegativities, bond dissociationenthalphies and E^0 values (SRP) are in the same order.

Statement-B: Xenon fluorides readily hydrolyses and the hydrolysis reactions of all xenon fluorides are redox reactions.

In the light of the above statements, choose the correct answer from the options given below.

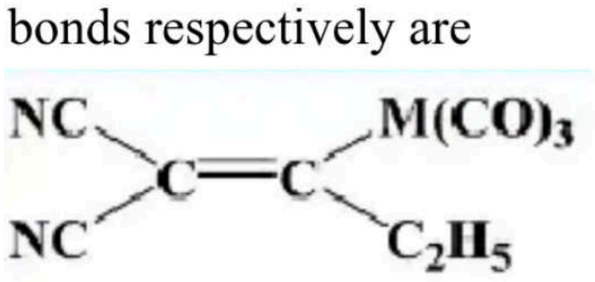
(1) Both statements A and B are true

(2) Statement A is false but statement B is true

(3) Both statements A and B are false

(4) Statement A is true but statement B is false

50. In a compound the number of sigma and pi bonds respectively are



(1) 19, 11 (2) 19, 5

(3) 13, 11 (4) 7, 3

51. The electronic configuration of four elements are

(I) $[Kr]5s^1$

(II) $[Rn]5f^{14}\,6d^1\,7s^2$

(III) $[Ar]3d^{10}\,4s^2\,4p^5$

(IV) $[Ar]3d^6\,4s^2$

Consider the following statements

(i) I shows variable oxidation state

(ii) II is a d-block element

(iii) The compound formed between I and III is covalent

(iv) IV shows single oxidation state

Which statement is true (T) or false (F)

(1) F T F F (2) F T F T

(3) F F T F (4) F F F F

52. Which of the following possess two lone pair of electrons on the central atom and square planar in shape

(I) SF_4 (II) XeO_4

(III) XeF_4 (IV) ICl_4^-

(1) I, III (2) II, IV

(3) III, IV (4) All

53. The correct order of increasing bond angles is

(1) $OF_2 < ClO_2 < H_2O < Cl_2O$

(2) $OF_2 < H_2O < Cl_2O < ClO_2$

(3) $OF_2 < H_2O < ClO_2 < Cl_2O$

(4) $ClO_2 < OF_2 < H_2O < Cl_2O$

54. Given below are two statements Assertion (A) and Reason (R).

Assertion-(A): From Titanium to Copper significant increase in the density is noticed in 3d series.

Reason-(R):The decrease in metallic radius coupled with increase in atomic mass results in general increase in the density of these elements.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.

(2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'

(3) 'A' is true and 'R' is false

(4) A is false, but R is true

55. Identify the incorrect match.

Name	IUPAC official name
A) Unnilunium	i) Mendelevium
B) Unniltrium	ii) Lawrencium
C) Unnilhexium	iii) Seaborgium
D) Unununnium	iv) Darmstadtium

(1) B, ii (2) C, iii

(3) D, iv (4) A, i

56. Correct statements among the following is/are

(i) In actinides Np, Pu, exhibits maximum oxidation states.

(ii) Actinides with cold water gives a mixture of oxide and hydride.

(iii) Ionisation enthalpies of early lanthanides are lower than ionisation enthalpies of early actinides.

(iv) Most of actinides are slightly affected by HNO_3 due to protective oxide layer formation.

(1) i, ii, iii are correct

(2) ii, iii are correct

(3) i, iv are correct

(4) i, ii, iii, iv are correct

57. Which of the following cation is precipitated first as hydroxide when NH_4OH solution is added drop by drop?

(1) Ca^{2+} (2) Ba^{2+}

(3) Sr^{2+} (4) Al^{3+}

58. The reaction of and follows the equation

$$A_{2(g)} + B_{2(g)} \rightarrow 2AB_{(g)}$$

The following data were observed

$[A_2]_0$	$[B_2]_0$	Initial rate of appearance of AB(g) (in Ms^{-1})
0.10	0.10	2.5×10^{-4}
0.20	0.10	5×10^{-4}
0.20	0.20	10×10^{-4}

The value of rate constant for the above reaction is

(1) 2.5×10^{-4} (2) 2.5×10^{-2}

(3) 1.25×10^{-2} (4) 1.25×10^{-4}

59. A first order reaction is 50% completed in 20 minutes at 27^0C and in 5 minutes at 47^0C. The energy of activation of the reaction is

(1) 43.85 kJ/mol (2) 55.14 kJ/mol

(3) 11.97 kJ/mol (4) 6.65 kJ/mol

60. Match the list – I with list – II.

List – I (Solid salt treated with dil H_2SO_4)	List – II (Anion detected)
A) Effervescence of colourless gas	I) NO_2^-
B) Gas with smell of rotten egg	II) CO_3^{2-}
C) Gas with pungent smell	III) S^{2-}
D) Brown fumes	IV) SO_3^{2-}

Choose the correct answer from the options given below:

(1) A – II, B – III, C – IV, D – I

(2) A – IV, B – III, C – II, D – I

(3) A – I, B – II, C - III, D – IV

(4) A – II, B – III, C – I, D – IV

61. The boiling points elevation constant for toluene is $3.32 Kkgmol^{-1}$. The normal boiling point of toluene is 110.7^oC. The enthalpy of vaporisation of toluene would be nearly

(1) $17.0\ kJmol^{-1}$ (2) $34.0\ kJmol^{-1}$

(3) $51.0\ kJmol^{-1}$ (4) $68.0\ kJ\,mol^{-1}$

62. Complexes formed in the following methods are

(I) Mond's process for purification of nickel

(II) Removal of unreacted AgBr from photographic plate

(III) Removal of lead poisoning from the body

(1) I – $Ni(CO)_4$, II–$[Ag(CN)_2]^-$, III – $[Pb(EDTA)]^{2-}$

(2) I – $Ni(CO)_4$, II–$[Ag(S_2O_3)_2]^{3-}$, III – $[Pb(EDTA)]^{2-}$

(3) I – $Ni(CO)_6$, II–$[Ag(S_2O_3)_2]^{3-}$, III – $[Pb(EDTA)]^{4-}$

(4) I – $Ni(CO)_6$, II–$[Ag(S_2O_3)]^-$, III – $[Pb(EDTA)]^{2-}$

63. For the following reaction, pick out the incorrect interpretations?

$$2A_{(g)} + B_{(g)} \rightarrow 4C_{(g)} + D_{(g)}$$

(1) 2 moles of A and 1 mole of B reacts to form 4 moles of C and 1 mole of D

(2) 2 gm of A and 1 gm of B reacts to give 4gm of C and 1 gm of D

(3) 2 molecules of A and 1 molecule of B reacts to give 4 molecules of C and 1 molecule of D

(4) 2 vol. of A and 1 vol. of B reacts to give 4 vol. of C and 1 vol. of D (At STP)

64. Suppose two elements X and Y combine to form two compounds XY_2 and X_2Y_3 when 0.05 mole of XY_2 weighs 5g while 3.011×10^{23} while molecules of X_2Y_3 weighs 85g. The atomic masses of X and Y are respectively

(1) 20, 30 (2) 30, 40

(3) 40, 30 (4) 80, 60

65. What will be the orbital angular momentum of an electron in 2s-orbital?

(1) Zero (2) One

(3) Two (4) Three

66. Time taken for an electron to complete one revolution in Bohr orbit of hydrogen atom is

(1) $\frac{4\pi^2mr^2}{nh}$ (2) $\frac{nh}{4\pi^2mr^2}$

(3) $\frac{2\pi mr}{n^2h^2}$ (4) $\frac{h}{2\pi mr}$

67. Dissolution of ammonium chloride in water is an endothermic reaction, yet it is a spontaneous process. This is due to the fact that

(1) ΔH is $+ve$, ΔS is $-ve$

(2) ΔH is $-ve$, ΔS is $+ve$

(3) ΔH is $+ve$, ΔS is $+ve$ and $\Delta H < T\Delta S$

(4) ΔH is $+ve$, and $\Delta H > T\Delta S$

68. Dissociation of water takes palce in two steps.

$$H_2O \rightarrow H + OH;\ \Delta H = +497.8kJ$$

$$OH \rightarrow H + O;\ \Delta H = 428.5kJ$$

What is the bond energy of $O-H$ bond?

(1) $463.15kJmol^{-1}$ (2) $428.5kJmol^{-1}$

(3) $69.3\ kJ\,mol^{-1}$ (4) $926.3\ kJ\,mol^{-1}$

69. Given below are two statements Assertion (A) and Reason (R).

Assertion-(A): For the reaction at certain temperature $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)}$

there will be no effect by addition of inert gas at constant volume.

Reason-(R):Moalr concentration of all gases remains constnat.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.

(2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'

(3) 'A' is true and 'R' is false

(4) A is false, but R is true

70. Aqueous solution of $Al_2(SO_4)_3$ is

(1) Strong acid (2) Weak acid

(3) Strong base (4) Weak base

71. What will be the pH of an aqueous solution of 1.0M ammonium formate?

Given: $pK_a = 3.8$ and $pK_b = 4.8$

(1) 7.5 (2) 3.4

(3) 6.5 (4) 10.2

72. The reaction which take palce at cathode in Leclanche cell is

(1) $HgO + H_2O + 2e^- \rightarrow Hg + 2OH^-$

(2) $Zn \rightarrow Zn^{2+} + 2e^-$

(3) $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$

(4) $PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$

73. If $E^{o}_{Au^{+}/Au}$ is 1.69 V and $E^{o}_{Au^{3+}/Au}$ is 1.40 V then $E^{o}_{Au^{+}/Au^{3+}}$ will be

(1) 0.19 V (2) 2.945 V

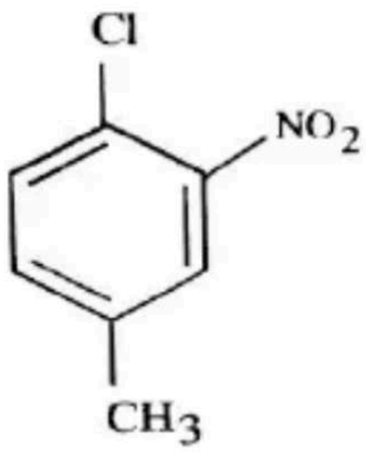
(3) 1.255 V (4) −1.255V

74. The weight ratio of Al and Ag deposited using the same quantity of current is

(1) 9 : 108 (2) 2 : 12

(3) 108 : 9 (4) 3 : 8

75. The IUPAC name of



(1) 1–Chloro–2–nitro–4– methylbenzene

(2) 1–Chloro–4–methyl–2– nitrobenzene

(3) 2–Chloro–1–nitro–5– methylbenzene

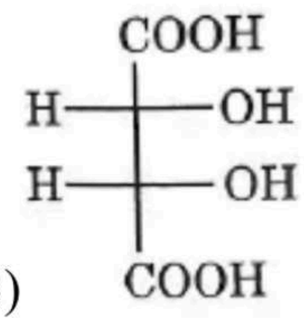
(4) m-Nitro-p-chlorotoluene

76. Which of the following will not show optical isomerism?

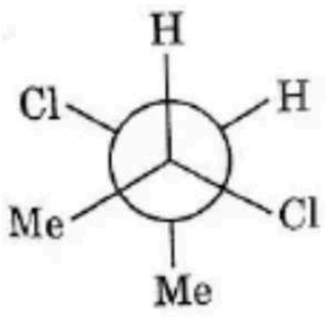
(1) $Cl-CH=C=C=CH-Cl$

(2) $Cl-CH=C=C=C=CH-Cl$

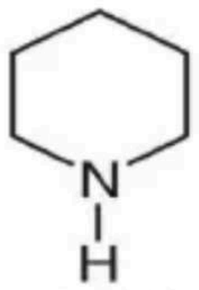
(3)



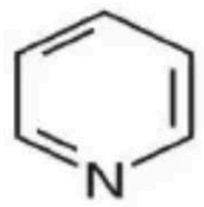
(4)



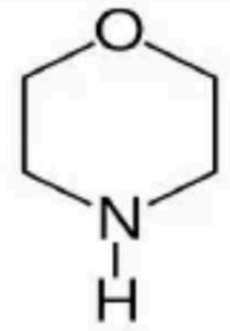
77. Select the basic strength order of following molecules?



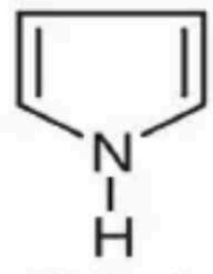
(I) Piperidine



(II) Pyridine



(III) Morpholine



(IV) Pyrrole

(1) IV > I > III > IV

(2) III > I > IV > II

(3) II > I > III > IV

(4) I > III > II > IV

78. Which of the following acids on decarboxylation gives isobutane

(1) 2, 2-Dimethyl butanoic acid

(2) 2,2-Dimethyl propanoic acid

(3) 3-Methyl pentanoic acid

(4) 2-methyl butanoic aicd

79. Given below are two statements Assertion (A) and Reason (R).

Assertion-(A): Boiling point of pentane is higher than 2,2-dimethylpropane.

Reason-(R): There is steady increase in boiling point with increase in molecular mass.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.

(2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'

(3) 'A' is true and 'R' is false

(4) A is false, but R is true

80. S_{N^1} reaction is fastest in

(1) CH_3CH_2Br

(2) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_3$

(3) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-Cl$

(4) $CH_3-\underset{\underset{CH_2}{|}}{\underset{\underset{CH_3}{|}}{CH}}-Cl$

81. Which of the following reactions does not take place?

(1) $C_2H_5Br + KNO_2 \rightarrow C_2H_5 - O - N = O + KBr$

(2) $C_2H_5Br + AgNO_2 \longrightarrow C_2H_5 - N\begin{matrix}=O\\ \diagdown O\end{matrix} + AgBr$

(3) $C_2H_5Br + AgCN \rightarrow C_2H_5NC + AgBr$

(4) $C_2H_5Br + KCN \rightarrow C_2H_5NC + KBr$

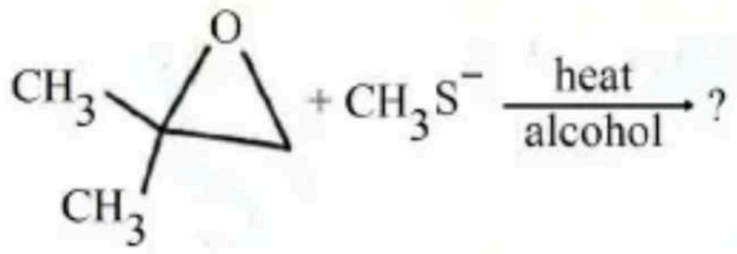
82. Consider the following reaction,

$$\text{Phenol} \xrightarrow{\text{Zndust}} X \xrightarrow[\text{anhyd.}AlCl_3]{CH_3Cl} Y \xrightarrow{\text{alkaline } KMnO_4} Z.$$ The product Z is

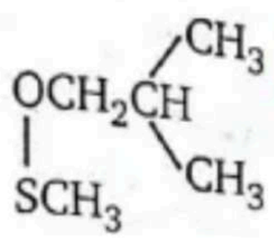
(1) benzaldehyde (2) benzoic acid

(3) benzene (4) toluene

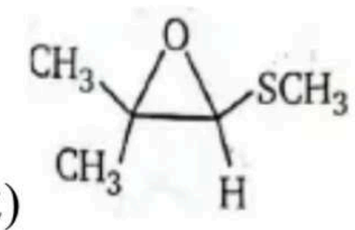
83. What is the major product of the following reaction?



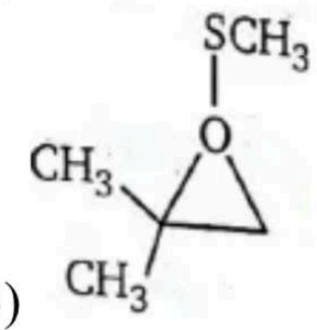
(1)



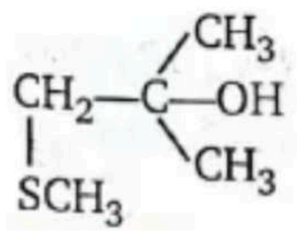
(2)



(3)

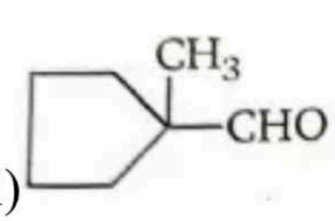


(4)

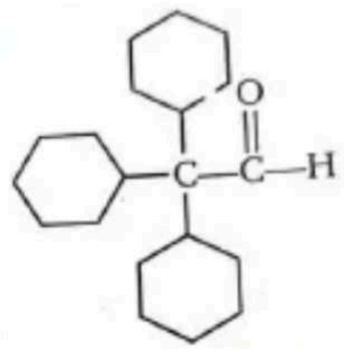


84. Which of the following compound can not show aldol condensation reaction in basic medium?

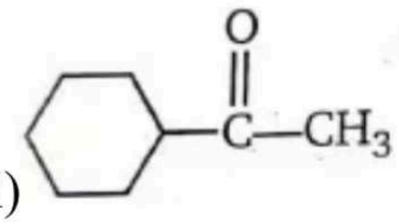
(i)



(ii)



(iii)



(iv) $H - \overset{\overset{O}{\|}}{C} - H$

(1) i & iii (2) i, ii & iv

(3) iii & iv (4) i, ii & iii

85. $H - \overset{\overset{O}{\|}}{C} - D \xrightarrow{OH^{\Theta}}$ product of Cannizzaro.

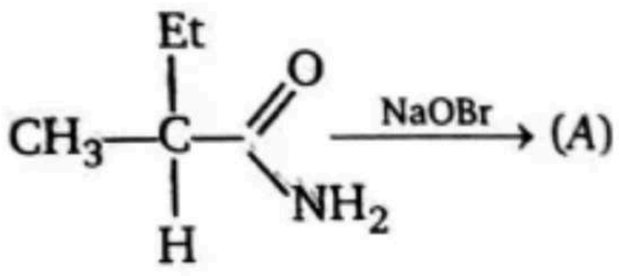
(1) $D - COO^{\Theta} + CH_2 - OD$ (D below CH$_2$)

(2) $H - COO^{\Theta} + D - COO^{\Theta}$

(3) $D - COO^{\Theta} + \underset{\underset{D}{|}}{\overset{\overset{D}{|}}{CH}} - OH$

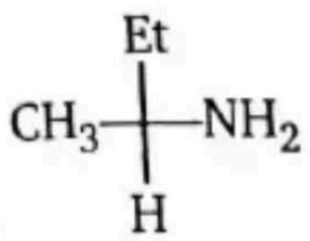
(4) $D - COO^{\Theta} + CH_2 - OH$ (D below CH$_2$)

86.

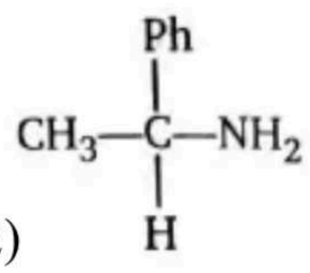


Major product of the reaction.

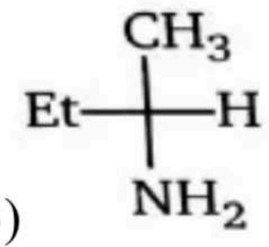
(1)



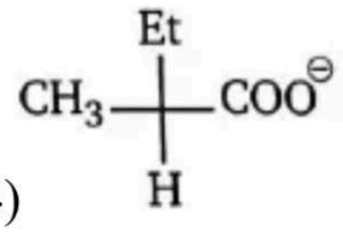
(2)



(3)



(4)



87. The reaction of chloroform with alcoholic KOH and ρ-Toluidine forms:

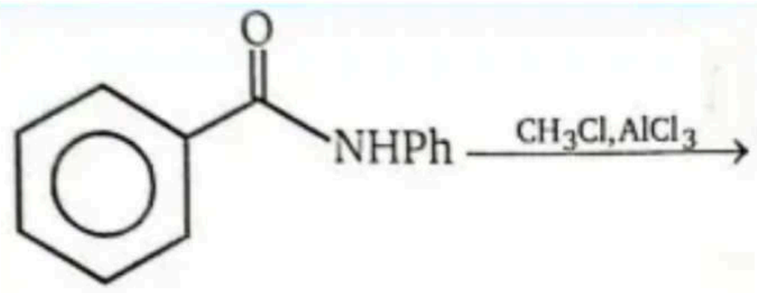
(1)

(2)

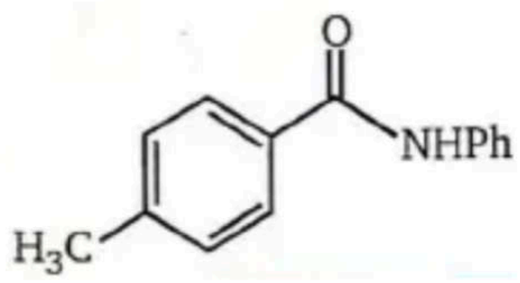
(3)

(4)

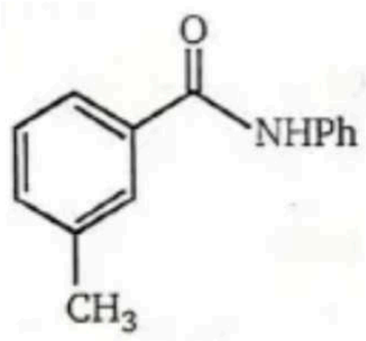
88. Give the major product of the following reaction



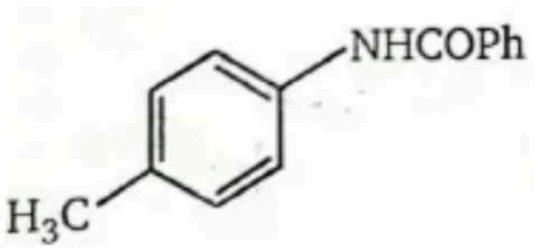
(1)



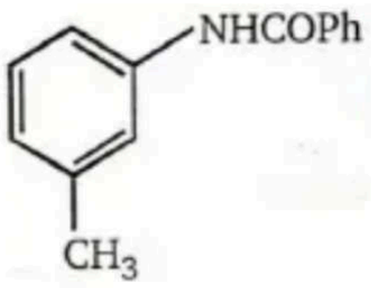
(2)



(3)



(4)



89. Given below are two statements Assertion (A) and Reason (R).

Assertion-(A): Purine bases present in DNA are adenine and guanine.

Reason-(R): the base thymine is present in RNA while base uracil is present in DNA.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.

(2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'

(3) 'A' is true and 'R' is false

(4) A is false, but R is true

90. Which of the following is true about the complex ?

(1) It exhibits geometrical isomerism

(2) It is paramagnetic complex

(3) Its geometry is tetrahedron

(4) Platinum is sp^3 hybridised

BIOLOGY

91. Match the following

List – I	List – II
I) Human beings	A) Legume fruit, Mendel's experimental plant
II) Mycoplasma	B) Budding, alcoholic fermentation
III) Pea	C) Self consciousness, multiple allelism in blood groups
IV) Yeast	D) Smallest living cells, wall less monerans

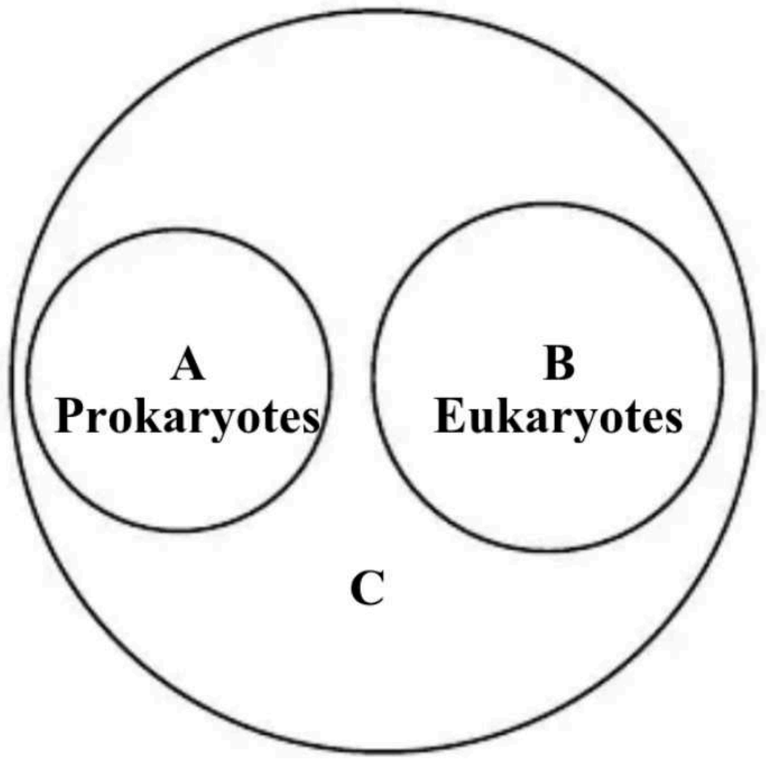
(1) I – A, II – B, III – C, IV – D

(2) I – C, II – D, III – A, IV – B

(3) I – D, II – C, III – B, IV – A

(4) I – C, II – B, III – D, IV – A

92. Observe the following diagram representing organisms and choose the correct option that represents 'C'



(1) Kelps (2) Viruses

(3) Fungi (4) Plants

93. Given below are two statements:

Assertion (A) : The hag fish is a cyclostome and the dog fish is a gnathostome, yet both of them are grouped under the same phylum.

Reason (R) : All chordates are not vertebrates.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

94. PCR and *invivo* DNA replication differ from one another in

I) Polymerisation/chain extension

II) Addition of primers

III) Removal of primers

IV) Discontinuous synthesis of one strand

V) Use of thermostable enzyme

(1) I and II only

(2) III, IV and V only

(3) I, IV and V only

(4) II and IV only

95. From the organisms, given below select the odd one on the basis of symmetry exhibited by them.

(1) *Locusta* (2) *Pavo*

(3) *Taenia* (4) *Meandrina*

96. Match the following

List – I	List – II
I) Totipotency	A) Progeny plants that are genetically identical to the original parent plant obtained through tissue culture
II) Micro propagation	B) Plants obtained by fusing protoplasts of two different plant varieties
III) Somaclones	C) Capacity to regenerate a whole plant from a cell/explant
IV) Somatic hybrids	D) Attaining self sufficiency in food grain production
V) Green revolution	E) Production of thousands of plants through tissue culture

(1) I – A, II – C, III – E, IV – D, V – B

(2) I – C, II – D, III – A, IV – E, V – B

(3) I – C, II – E, III – A, IV – B, V – D

(4) I – E, II – B, III – A, IV – D, V – C

97. *Rana tigrina*

a) can change its body colour as per the needs

b) can change its body temperature as per the changes in the external temperature

c) can perform branchial respiration in water and only pulmonary respiration on land

d) can exhibit hibernation in aquatic habitat and aestivation in terrestrial habitat

In the above the correct statement

(1) a only

(2) a & b only

(3) a, b & c only

(4) a, b, c & d

98. Read the following statements and choose the correct ones

A) Tapetum is the innermost layer surrounding a typical microsporangium

B) Cells of the nucellus have abundant reserve food materials

C) A typical angiosperm embryosac, at maturity is 8 celled 7 nucleated

D) Geitonogamy is genetically similar to autogamy

E) If the female parent produces bisexual flowers emasculation is done

F) Embryo development precedes endosperm development

(1) A, B, D, E (2) B, C, D

(3) B, C, E, F (4) A, C, F

99 Match the following

List – I	List – II
I) Algin, carrageen	A) Bacteria
II) Sexual reproduction occurs by direct DNA transfer between cells	B) Apomictic grasses
III) Deviation from sexual reproduction resulting in formation of seeds with embryo	C) Hydrocolloids obtained from algae
IV) Multiplication always occur inside the host cells	D) Viruses

(1) I – A, II – B, III – C, IV – D

(2) I – B, II – C, III – D, IV – A

(3) I – D, II – C, III – B, IV – A

(4) I – C, II – A, III – B, IV – D

100. India share of global species diversity is

(1) 14 % (2) 11 %

(3) 8.1 % (4) 2.4 %

101. Chlorophyll '*a*' and '*b*' are present in

A) Cyanobacteria

B) Brown algae

C) *Euglena*, higher plants

D) Green algae

(1) A and B (2) A and C

(3) C and D (4) B and C

102. All are located in/on thylakoid membrane except

(1) Enzymes of dark phase

(2) ATP synthase

(3) Photosynthetic pigments

(4) Plastoquinone

103. How many of the structures given below are diploid and represent sporophytic phase ?

Zygote, Antipodal cell, Nucellus, Pollen mother cell, Synergid, Primary endosperm cell

(1) Four (2) Three

(3) Five (4) Six

104. Urea synthesized in the liver must reach the kidney for its elimination for this purpose the urea must pass through which of the following blood vessels?

(1) Hepatic artery and renal vein

(2) Renal artery and hepatic portal vein

(3) Renal portal vein and hepatic artery

(4) Hepatic vein and renal artery

105. The number of nephrons present in human being is nearly

(1) 1 lakh (2) 2 lakh

(3) 10 lakh (4) 20 lakh

106. Match the following columns

Column – I		Column – II	
A.	Lysozyme	I.	CO_2 transport
B.	Renin	II.	Bactericidal
C.	Carbonic anhydrase	III.	Muscle contraction
D.	Myosin ATPase	IV.	Kidney autoregulation

Choose the correct answer from the options given below.

(1) A – II, B – IV, C – I, D – III

(2) A – I, B – II, C – III, D – IV

(3) A – II, B – IV, C – III, D – I

(4) A – II, B –I, C – III, D – IV

107. Which of the following structure is actively involved in the synthesis of *r*RNA ?
(1) Plasmid (2) Nucleolus
(3) Ribosome (4) Centriole

108. Given below are two statements:
Assertion (A) : Pyramid of energy is inverted in a parasitic food chain.
Reason (R) : In parasitic food chain the number of organisms increases and the size of organisms decreases from lower trophic level to the higher trophic level.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

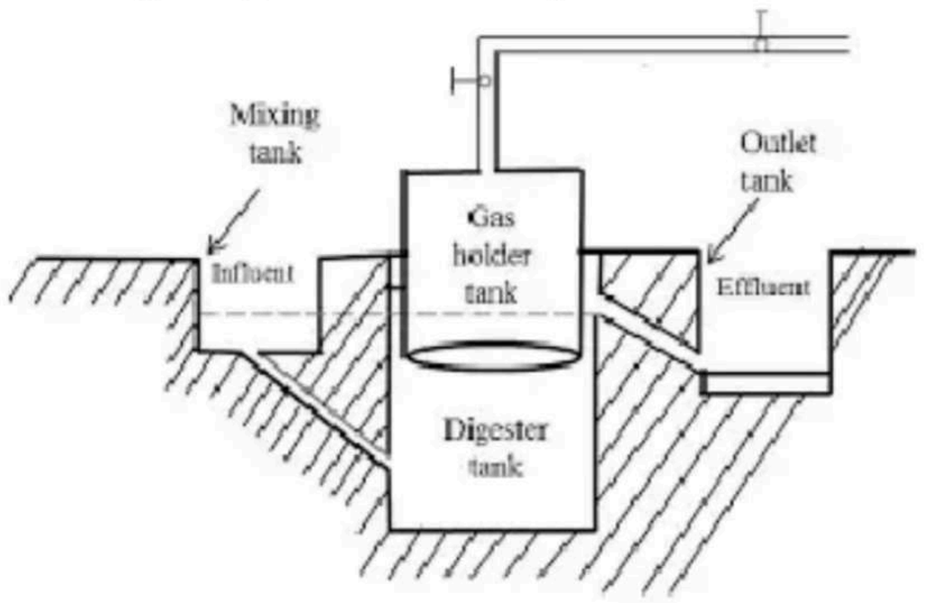
109. Assertion (A) : Chromosomes of a bivalent start repelling from one another in diplotene
Reason (R) : Synaptonemal complex dissolves in diplotene
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true and R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

110. Which of following structures are not made up of same tissue?
(1) Epiglottis and earlobe
(2) Pleural membrane and pericardial membrane
(3) Metacarpal and metatarsal
(4) Intervertebral discs and intercalated discs

111. α – thalassemia a genetical disorder which requires periodic transfusion of blood is controlled by
(1) One gene located on 16th chormosome
(2) One gene located on 11th chormosome
(3) Two genes located on 16th chormosome
(4) Two genes located on 11th chormosome

112. Statement – I : Plants exhibit continuous (indeterminate) growth throughout their life cycle
Statement – II : Plants possess meristematic tissues that help in growth
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

113. The figure given below represents



(1) Fermentor
(2) Sewage treatment plant
(3) Biogas plant
(4) Bioreactor

114. Which of the following are correct about the human respiratory system?
I) The trachea and bronchi are supported by complete circles of cartilage
II) The recoil of elastic fibres surrounding the alveoli helps to move air out during expiration
III) The walls of the alveoli are made up of simple cuboidal epithelium
IV) The initial bronchioles are supported by incomplete cartilaginous rings.
Choose the correct answer from the options given below
(1) I and II (2) I and III
(3) II and IV (4) III and IV

115. The members of the largest phylum are the first to have
(1) Three embryonic germ layers
(2) Adapted their life completely on land
(3) Special organs for respiration
(4) The capacity of regeneration of damaged organs.

116. Casparian strips occur in
(1) Epidermis of anther
(2) Pericycle of stem
(3) Endodermis of root
(4) Bundle sheath of monocots

117. Match the following columns

Column – I		Column – II	
A.	Epinephrine	I.	Peptide hormone
B.	T_4	II.	Steriod hormone
C.	Progesterone	III.	Iodothyronine
D.	Glucagon	IV.	Amino acid derivative

Choose the correct answer from the options given below.
(1) A – IV, B – III, C – I, D – III
(2) A – III, B – IV, C – I, D – II
(3) A – III, B – IV, C – II, D – I
(4) A – IV, B –III, C – II, D – I

118. Plasmodesmata are
(1) Protein filaments of cytoskeleton
(2) Phospholipids in cell membrane
(3) Cytoplasmic strands between adjacent plant cells
(4) Microbodies in a cell

119. Choose the character unrelated to *Drosophila* (fruit fly)
(1) XY sex chromosomes present
(2) Body colour and eye colour genes are on X chromosome
(3) Life cycle is for two years
(4) Possess giant polytene chromosomes in salivary glands of some species

120. Centres which control respiration, cardiovascular reflexes and gastric secretion are located in
(1) Fore brain (2) Mid brain
(3) Hind brain (4) Spinal cord

121. Given below are two statements:
Statement – I : Synovial joints are freely movable joints.
Statement – II : Synovial joints are characterized by fluid filled synovial cavity between articulating bones.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Only statement I – is correct, statement – II is incorrect
(4) Only statement I – is incorrect, statement – II is correct

122. Assertion (A) : Mitochondria are considered as "power house" of the cell
Reason (R) : ATP is synthesised in mitochondria due to aerobic respiration are available for cell functions.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true and R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

123. The factors which help in the process of spermiogenesis are secreted by
(1) Leydig cells
(2) Sertoli cells
(3) Follicular cells
(4) Granulosa cells

124. Electron transport occurs in all the following regions except
(1) Plasma membrane of aerobic bacteria
(2) Mitochondrial cristae
(3) Membrane of ER
(4) Thylakoid membrane

125. Out of the following diseases which are caused due to bacterial infection?

a) Typhoid b) Elephantiasis

c) Cholera d) Tuberculosis

(1) a and b only (2) b and c only

(3) a, c and d only (4) a, b, c and d

126. Choose the incorrect statement

(1) Male parent is pollen donor in artificial hybridization

(2) PQ is mobile hydrogen carrier

(3) Water is the electron donor to PS II in non cyclic electron transport

(4) NADH is electron and proton donor to dark phase

127. Australia is considered as land of Marsupialia. The probable reason for the survival and success of pouched mammls in Australia is

(1) Plenty of food availability

(2) Suitable living habitats

(3) Maximum reproductive potential

(4) Absence of compeitation from other mammals

128. Match the following columns

Column – I		Column – II	
A.	Reserpine	I.	Over explotation
B.	Western ghats	II.	Alien species
C.	*Clarias gariepinus*	III.	Rauwolfia
D.	Passenger pigeon	IV.	Greater amphibian diversity

Choose the correct answer from the options given below.

(1) A – III, B – IV, C – II, D –I

(2) A – III, B – IV, C – I, D – II

(3) A – IV, B – III, C – II, D – I

(4) A – IV, B –III, C – I, D – II

129. Choose the incorrect match in relation to events and processes

(1) Electrophoresis : Separation of DNA fragments

(2) Elution : Separation of DNA fragments from the agarose gel

(3) PCR : *invitro* DNA replication

(4) Activated sludge : Sediment formed in primary settling tank

130. Herbivores

a) are primary consumers

b) are generally plant eating

c) occupy first trophic level of a food chain

d) have stored energy called NPP

In the above the correct statement are

(1) a only (2) a and b only

(3) a, b and d only (4) a, b, c and d

131. All possess false fruits along with true fruit except one. What is it ?

(1) Cashewnut (2) Coconut

(3) Strawberry (4) Apple

132. Match the following columns

Column – I		Column – II	
A.	Primates	I.	Superclass
B.	Pisces	II.	Class
C.	Agnatha	III.	Order
D.	Cyclostonata	IV.	Division

Choose the correct answer from the options given below.

(1) A – II, B – III, C – IV, D – I

(2) A – III, B – I, C – II, D – IV

(3) A – III, B – I, C – IV, D – II

(4) A – IV, B –III, C – II, D – I

133. A dioecious plant with thalloid body is _____, a dioecious plant with naked seeds is _____ respectively

(1) *Marchantia*, *Cycas*

(2) *Funaria, Pinus*

(3) *Marchantia,* Maize

(4) *Sphagnum, Zostera*

134. Pumping of calcium ions into cisternae of sarcoplasmic reticulum results in
(1) Formation of cross bridges
(2) Binding of myosin head with actin
(3) Pulling of acin attached to z - membrane
(4) Masking of active sites on actin filaments

135. Internal and external reasons for high yield in a sugarcane plant are due to _____ and _____ respectively
(1) C_3 cycle, more CO_2 in atmosphere
(2) Photorespiration, bolting
(3) C_4 cycle, gibberellic acid treatment
(4) C_3 cycle, CO_2 enriched atmosphere

136. Given below are the dinosaurs and their features exhibited by them. Find the correct.
A) *Pteranodon* – Probably flying
B) *Triceratops* – Horns projected on head
C) *Stegosaurus* – Middorsal crest
D) *Brachiosaurus* – Prolonged neck and tail
(1) A only (2) A and B only
(3) A, B and C only (4) A, B, C and D

137. In *lac* operon expression of this gene occurs independent of lactose in *E.coli.* What is it ?
(1) *z* gene (2) *i* gene
(3) *y* gene (4) *a* gene

138. Conjuctive tissue is found in the anatomy of
(1) stem (2) root
(3) ovule (4) anther

139. Identify the incorrect from the following
(1) Epicardium, myocardium, endocardium – heart wall
(2) Perimetrium, myometrium, endometrium – Uterus wall
(3) Zona glomerulosa, zona fasiculata, zona reticularis – Adrenal medulla
(4) Parsdistalis, parsintermedia, parsnervosa – Hypophysis

140. Home use kits for determining a women's pregnancy depend on the detection of one hormone in the her urine. This hormone is
(1) L.T.H (2) L.H
(3) Progesterone (4) hCG

141. Given below are two statements:
Assertion (A) : Stem cells of innercell mass have pleuropotency
Reason (R) : Cells of innercell mass differentiates into outer ectoderm, middle mesoderm and inner endoderm which give rise to tissues and organs.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

142. Cell organelles associated with the process of lipid synthesis, protein synthesis and carbohydrate synthesis respectively are
(1) Smooth ER, ribosome, chloroplast
(2) Rough ER, golgi complex, mitochondria
(3) Chloroplast, peroxisome, smooth ER
(4) Glyoxysome, rough ER, chloroplast

143. Choose the structure that does not have proteins
(1) Cytoskeleton
(2) Pellicle in *Euglena*
(3) Nucleosome
(4) Volutin granules

144. Vasectomy has no effect on the sexual life of male because
(1) Vesectomy affects the hormonal secretion but the person can ejaculate semen with number of sperms
(2) Vasectomy doesnot affect the sexual desire but can cause (erectile dysfunction) impotency
(3) Vasectamised male can still ejaculate and has normal secretions of sex hormones in blood
(4) Vasectomized male cannot ejaculate semen but the sex hormone levels are normal

145. The counter current mechanism
A) Helps to maintain concentration gradient in the medullary interstitium
B) Is unique to mammalian excretory physiology
C) Is made possible by cortical nephrons
D) Is possible due to special arrangement of Henle's loop and vasarecta
Choose the correct answer from the options given below
(1) A, B, C and D (2) A, C and D only
(3) A, B and C only (4) A, B and D only

146. Svedberg unit is a measure used for _____ and centimorgan is a measure used for _____ respectively
(1) RNA, Nucleus
(2) Viruses, Cell membrane
(3) Ribosome, Gene distance
(4) Golgi complex, Gene length

147. The total volume of air a person can inspire after normal expiration, includes
(1) TV + IRV (2) TV + ERV
(3) ERV + RV (4) ERV + TV + IRV

148. Restriction endonuclease breaks ____bonds and helicase breaks ___ bonds
(1) Hydrogen, ester
(2) Phosphodiester, hydrogen
(3) Peptide, hydrogen
(4) Glycosidic, hydrogen

149. Prokaryotic cell generally possess
A) Cell membrane
B) Membrane bound cell organelles
C) Ribosomes
D) Genetic material
E) Cell wall
F) Cilia
(1) All except B, F (2) All except D, E
(3) All except B, E (4) All except A, D

150. Blood exhibits coagulation in response to an injury. This mechanism involves series of reactions involving several factors. Which of the following is not readily present in blood plasma?
(1) Calcium (2) Fibrinogen
(3) Prothrombin (4) Thrombokinase

151. Match the following

List – I	List – II
I.Phyllotaxy	A.Externally visible trait
II.Phenotype	B.Disease caused by smallest moneran
III.Pleuro pneumonia	C.Type of cohesion in stamens
IV.Polyadelphy	D. Arrangement of leaves on the stem/branch

(1) I – B, II – C, III – A, IV - D
(2) I – C, II – D, III – A, IV - B
(3) I – D, II – A, III – B, IV – C
(4) I – A, II – B, III – C, IV – D

152. High levels of species richness and high degree of species endemism can be observed in the following
(1) Sacred grooves of khasi and Jaintia hills
(2) Western Ghats and Srilanka biodiversity hotspot
(3) Nanda devi biosphere reserve
(4) Kaziranga national park

153. Insulin is synthesized as prohormone in
(1) Humans only
(2) Mammals only
(3) Mammals other than humans
(4) Mammals including humans

154. Statement – I : Tryptophan is an aromatic amino acid.
Statement – II : Exception to the property "code is degenerate" is applicable to codon that codes for tryptophan
(1) Both statement I and statement II are correct.
(2) Both statement I and statement II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

155. Given below are two statements:

Statement – I : Sudden damage of heart muscle by inadequate blood supply is heart attack.

Statement – II : Sudden stoppage of the beating of the heart is heart arrest.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Only statement I – is correct, statement – II is incorrect

(4) Only statement I – is incorrect, statement – II is correct

156. Choose the incorrect match among the following

Phenomenon		Character
(1) Linkage	:	Deviation from dihybrid test cross ratio
(2) Pleiotropic genes	:	Possess more than two alleles that determine a single phenotype
(3) Polygenic inheritance	:	Phenotype is spread across a gradient in population and is influenced by environment
(4) Incomplete dominance	:	F_2 phenotypic ratio is 1 : 2 : 1

157. Given below are the characteristics of the animals commonly known as "Comb jellies".

a) They have eight ciliated comb plates for locomotion

b) They are exclusively aquatic

c) They are capable of emission of light

d) They reproduce only by asexual means

e) Their life cycle includes larval stage

How many of the above are correct

(1) 5 (2) 4

(3) 3 (4) 2

158. Arrange the following male reproductive organs of frog in correct sequence as per the passage of spermatozoa during pseudocopulation.

a) Vasa efferentia

b) Urinogenital duct

c) Testes

d) Bidder's canal

e) Cloaca

(1) a – b – c – d – e

(2) c – d – a – b – e

(3) b – c – a – e – d

(4) c – a – d – b – e

159. Match the following

List – I	List – II
I. Proteinaceous pellicle	A. *t*RNA
II. Abnormally folded protein	B. *Euglena*
III. Nucleic acid folded like a clover leaf	C.Haemoglobin
IV. Quaternary structure of protein	D.Prion

(1) I – B, II – D, III – A, IV - C

(2) I – C, II – D, III – A, IV - B

(3) I – D, II – A, III – B, IV – C

(4) I – A, II – B, III – C, IV – D

160. Which of the following has fructose in it?

(1) Blood (2) Saliva

(3) Semen (4) Menstrual fluid

161. Choose the incorrect match from the following in relation to colour

(1) High rate of photosynthesis – Blue and red colour

(2) Visualisation of DNA in gel electrophoresis – Bright orange colour

(3) Transformed colonies screened using β gal marker gene with amp – Blue and white colour

(4) Fucoxanthin in brown algae – Red colour

162. Read the following statements regarding HIV virus

A) It is a type of retrovirus which has an envelope enclosing the RNA genome

B) It multiplies in helper T cells and produce progeny virus

C) In the macrophages of host, RNA genome of the virus replicates to form viral DNA

D) Reverse transciptase becomes inactive in the host cells

Choose the correct answer with all correct statements from the given below options

(1) A, B, C and D

(2) A, B and C only

(3) C and D only

(4) A and D only

163. Zygomorphic flower is seen in a genus namely

(1) Lily (2) *Datura*

(3) *Hibiscus* (4) *Dolichos*

164. Statement - I : Transgenic golden rice is rich in beta carotene

Statement - II : Transgenic tobacco resistant to nematode *Meloidogyne* has a gene introduced from a bacterium namely *Agrobacterium tumefaciens*

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

165. The picture given below is a prominent example for



(1) Commensalism (2) Parasitism

(3) Amensalism (4) Mutualism

166. Match the following columns

Column – I		Column – II	
A.	MTP act legalized	I.	1990
B.	Family planning programme is initiated	II.	1997
C.	Launch of human genome project	III.	1951
D.	Rosie, cow produced	IV.	1971

Choose the correct answer from the options given below.

(1) A – IV, B – III, C – I, D – II

(2) A – IV, B – III, C – II, D – I

(3) A – III, B – III, C – I, D – II

(4) A – III, B –IV, C – I, D – II

167. Given below are two statements:

Statement – I : It is not possible for the evolution of new species from preexisting species in present day environment.

Statement – II : Presente day atmosphere is oxydising and relatively with low temperature than the primitive earth.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I – is correct, statement – II is incorrect

(4) Statement I – is incorrect, statement – II is correct

168. Duplication of DNA without increase in chromosome number occurs during ___ phase in a cell

(1) Anaphase I (2) S phase

(3) Prophase I (4) Telophase

169. Read the following statements and choose the correct ones

A) Thymine is a nucleoside present in DNA

B) Carotenoids are accessory pigments present in stroma of chloroplast

C) Formation of acetyl Co.A from pyruvic acid involves removal of CO_2 from pyruvic acid

D) Formation of embryo within the female gametophyte retained over sporophyte is an event precursor to seed habit observed in pteridophytes

(1) A and B (2) B and C

(3) C and D (4) B and D

170. Which of the following diseases are transmitted only through sexual intercourse

(1) HIV and Hepatitis B

(2) HIV and Chlamydiasis

(3) Hepatitis B and Trichomoniasis

(4) Gonorrhoea and Syphilis

171. Choose the incorrect statement from the following

(1) Obliquely placed carpels are in members of Solanaceae

(2) Intercalary meristem is in monocots (grasses)

(3) RNA polymerase enzyme catalyses DNA unwinding in transcription

(4) Enzymes increase activation energy of substrate

172. Ascospores are haploid sexual meiospores observed in the life cycle of

(1) *Agaricus* (2) *Alternaria*

(3) *Neurospora* (4) *Rhizopus*

173. Polymorphism in DNA sequence is the basis of

(1) DNA finger printing only

(2) Genetic mapping only

(3) DNA fingerprinting and genetic mapping

(4) DNA sequencing only

174. Given below are two statements: about life cycle of malarial parasite.

Statement – I : Sporozoites are formed in female *Anopheles* mosquito and develop further in man.

Statement – II : Gametocytes are formed in man and further develop in female *Anapheles* mosquito

In the light of the above statements, choose the correct answer from the options given below:

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I – is correct, statement – II is incorrect

(4) Statement I – is incorrect, statement – II is correct

175. Match the following in relation to scientists and their experiments

Scientist	Experiment
A) Taylor	I) ^{14}C in CO_2
B) Meselson and Stahl	II) Radioactive thymidine
C) Calvin	III) ^{32}P and ^{35}S
D) Hershey and Chase	IV) Heavy ^{15}N in NH_4Cl

(1) A – IV, B – I, C – III, D - II

(2) A – II, B – IV, C – I, D – III

(3) A – III, B – III, C – II, D – I

(4) A – I, B – II, C – III, D – IV

176. Choose the correct match from the following

(1) *Atropa belladonna* – Brassicaceae

(2) *Allium cepa* – Liliaceae

(3) *Althea rosea* – Solanaceae

(4) *Artemesia vulgaris* – Malvaceae

177. Identify the incorrect from the following

(1) Surgical methods are also called sterilization methods.

(2) Emergency contraceptives are effective within 72 hours after ovulation

(3) Diaphragms, vaults and cervical caps are reusable.

(4) Oral pills inhibit ovulation and implantation.

178. Given below are two statements:

Assertion (A) : ELISA can enable very easy detection of infection through antigen antibody interaction.

Reason (R) : ELISA is a nucleic acid based diagnostic tool than can confirm presence of infectious microbe at earlystage.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

179. What is the net gain of ATP molecules for one glucose molecule in fermentation ?

(1) 8 (2) 4

(3) 2 (4) Zero

180. Which of the following structures move towards opposite poles in prophase ?

(1) Centrosome (2) Chromosome

(3) Mesosome (4) Ribosome